

Harnessing digital twin technology for enhanced aircraft turnaround efficiency

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ABSTRACT

The aircraft turnaround operation is a critical component of air transportation, traditionally carried out using manually coordinated devices. Consequently, the operational efficiency is often constrained by the volume of labor and individual capabilities. To address this, some airports are exploring the potential of automated turnaround operations, where flights are serviced by newly designed smart devices. This study proposes a unique method for predicting the efficiency of automated turnaround operations using a digital-twin model. First, we designed a sandbox-based apron to simulate the physical environment, as no automated apron currently exists. Next, we applied network planning technique to establish coordinated operation rules among the smart devices, creating an optimized procedure for aircraft automated turnaround operation. Our results indicate that, compared to the statistical efficiency of manual-device coordinated operations, the time required for an automated turnaround operation for a single flight can be reduced by approximately 24.53 %.

1. Introduction

The aircraft turnaround operation is the essential ground service which is provided by airports to ensure the smooth progress of civil aviation, and its procedure normally encompasses nearly 20 segments like boarding-bridge connection, cabin cleaning, refueling, and catering replenishment [44]. Practically, as the turnaround time directly impacts flight punctuality and significantly affects the overall operational continuity, stability, and economic viability of the airport, the efficiency of aircraft turnaround operation is the key indicator for both airline and airport operations. In this sense, how to improve the aircraft turnaround efficiency becomes an important problem in airport management, especially for hub airports [20].

Currently, most airports have implemented the manual-device method to conduct the aircraft turnaround operation. Although these transportation machineries such as flatbed trailers, passenger shuttle buses, platform vehicles, etc., have significantly improved the aircraft turnaround efficiency [36,37,39]. Their reliance on manual work may lead to the result that the current working efficiency heavily depends on the labor volume and their proficiency. Given that the number of flights in China is expected to increase continuously in the next 20 years [25],

while labor costs are likely to rise and the labor force may shrink in the long term [24], airports may confront the challenge of increasing workload but declining labor force [33]. Additionally, the frequent occurrence of severe weather in recent years (e.g., prolonged heat, extreme cold, high winds) has significantly affected the efficiency of the manual-device operation. Hence, some scholars and airports have proposed the concept of automated aprons, in which various smart devices would be created and utilized, such as smart passenger boarding bridges, intelligent tow trucks, automated trailers, etc. Steininger et al., [32].

Some domestic and foreign airports have begun to utilize smart devices in the aircraft turnaround operation. For instance, Chengdu Airport applied the automated boarding bridges in 2022 [17], and Singapore Airport utilized the automated luggage conveying system [6]. Such cases demonstrate that the automated turnaround operation, which is supported by smart devices, can significantly reduce the labor involvement and enhance the aircraft turnaround operation efficiency [9]. However, the adoption of such smart devices requires substantial investment. To be specific, the cost of a smart trailer is approximately 90 thousand yuan in RMB, a smart tow tractor costs about 25 thousand yuan, and the smart control systems may cost about 2 million yuan. Due to the huge initial costs, some airports have only invested in the smart

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devices for certain segments of the aircraft turnaround operation, but they refuse to further consider introducing more facilities until the efficiency of automated turnaround operation could be accurately evaluated [42].

The traditional simulations generally construct the aircraft turnaround process in a virtual environment based on the statistical data and the established rules. However, considering that the smart aprons have not been constructed yet, it is nearly impossible to collect and summarize the operational data or rules in reality. In the sense, comparison with the traditional simulation model, the sandbox-based apron module is created in our simulation with the purpose of constructing a physical version of the smart apron. As a result, the operational data about the smart devices on the apron can be collected in an experimental environment. In addition, the rules for aircraft turnaround operation using smart devices are quite different from those in current manual operation environment. For example, smart tugs are driven by multiple rules related to operational trajectories and timing, while manual tugs only rely on the operator's own judgment. Hence, the simulation model constructed in this paper sets up the rules specifically for smart devices during the aircraft automated turnaround operation.

Different from the traditional digital twin models, this study uses an apron sandbox in the model construction. The sandbox is a scaled-down apron with smart devices. On one hand, the smart devices can collect and transmit the operational data through radar, infrared, visual sensors, etc.; on the other hand, they are driven by designed algorithms for path planning, stepper motor control and so on. The aircraft automated turnaround operation process can be conducted on the sandbox-based apron under various scenarios in the laboratory, thus to test the feasibility and efficiency of the related smart support technologies.

It is evident that this study, by constructing a sandbox apron, has significantly reduced the development costs of the physical modules in traditional digital twin models. It has also expanded the range of scenarios applicable for simulation within the model. This approach is more conducive to scientifically verifying the feasibility of automated turnaround operation under different equipment configurations and scenarios.

The remainder of this paper is organized as follows: Section 2 provides a literature review, Section 3 establishes the digital twin model for the automated aircraft turnaround operation, Section 4 gives out the results, and Section 5 summarizes the research.

2. Literature review

Different from the turnaround operation with manual work, the automated turnaround operation highly depends on the development of smart devices. Some researchers have proposed various smart devices in the context of experiment as well as application. For example, Škultéty and Stalmašeková [29] designed a radio frequency identification system to detect aircraft emergency devices; Zhang et al. [41] developed the device for recognizing the moving target on the apron and Huang [11] proposed an unmanned apron support device. Such smart devices not only can collect the real-time data in terms of images, sounds, etc. through various sensors, they also have the ability to finish the turnaround operation automatically under the computer command [15]. The computer commands relevant to the autonomous operation of smart devices are composed of commands for single device as well as the ones for multi-device coordination. The commands about single-device are used to control the movements of one specific device, such as moving forward, rotating/extending mechanical arms and so on [28]. Moreover, the commands for multiple devices should be constructed incorporating the multi-device coordinated rules. The mentioned rules specify the enter and exit time of smart devices as well as their location at different time points [27]. Considering the single device control commands fall within the scope of mechanical research, we choose to focus on the analysis of multi-device coordinated rules.

The key of constructing the multi-device coordinated rules is to

optimize procedure of autonomous turnaround operation. For the turnaround operation with manual work, the procedure optimization is conducted based on the duration prediction of each segment within the turnaround operation [23]. On one hand, the procedure optimization for aircraft turnaround operation is explored. For instance, Sng and Hansman [30] optimized the segment sequence by constructing a petri net framework; Fan et al. [8] designed the procedure of aircraft turnaround operations based on island parallel algorithm, which realize the multi-line parallel scheduling. On the other hand, the prediction model is always developed using statistical data from practical operation. For instance, Kanable [13] developed a hybrid model integrating machine learning and video recognition technology to enhance the prediction accuracy of segment duration; Cui et al. [5] proposed a description model based on time transition petri net to improve the flight punctuality rate; Zhang et al. [40] proposed an enhanced sliding time window model to improve the prediction accuracy of the duration about the whole turnaround operation. Although the existing methods have significantly improved the efficiency of turnaround operation, they primarily focused on manual operation scenarios. Hence, the mentioned model and algorithm relevant to procedure construction, duration prediction and process configuration may be not suitable for the validating the operation efficiency of aircraft automated turnaround.

Owing to the lack of statistical data collected from smart aprons, our research tries to construct the digital twin model of aircraft automated turnaround operation in the laboratory. The model can be used to predict the duration of each segment and optimize the turnaround procedure, thus to verify the feasibility and efficiency of aircraft turnaround operation with smart device. Digital twin integrates multi-device interconnectivity [31], multi-dimensional simulation [19], data analysis and processing [10] as well as virtual reality [35]. It has been widely applied in multiple fields, such as workshop scheduling [43], product condition monitoring and fault diagnosis [26] and operation control of air transportation [7]. The digital twin models in current studies are generally composed of the physical module, the virtual module, the connection module, the data module and the service module [34]. The physical module is composed of various physical entities. The virtual module transfers the physical entities into digital ones in terms of one-to-one correspondence [18]. The connection module establishes the communication between physical and digital entities [12]. The data module stores various datasets, including sensor data, commands, operation rules, etc. Kong et al., [14]. The service module defines the operation rules for devices. The existing studies about airport management typically construct digital twin models based on practical production, the physical module is composed of the apron, terminal, devices, etc. in reality. However, considering that the smart aprons have not been constructed yet, the sandbox-based apron module must be incorporated into the digital twin model, thus to simulate the automated turnaround operation with smart devices.

The regulatory sandbox can be described as a controllable space in which the effects of various elements (e.g., products, services and policies) on the experiment outcomes can be tested [16]. The sandbox is initially applied in the field of financial technology and subsequently applied in a variety of research fields, such as healthcare [21], energy [1] and transportation [2]. Specifically, the existing studies have evaluated the impacts and benefits of prototype cars and traffic control strategies on the particularly designed sandbox [38]. However, the advantages of sandbox application in improving the aircraft turnaround efficiency have not received sufficient attention yet. We only find that Miltiadou et al. [22] designed a digital airport sandbox for collecting credible data in the digital transformation of air transportation. However, the airport sandbox proposed by them virtually simulate the airport operation but not conduct the virtual-physical linkages and not take into account the lack of smart device operation rules, so we should update the methods about constructing the sandbox of airport apron, thus to validate the efficiency of aircraft automated turnaround operation.

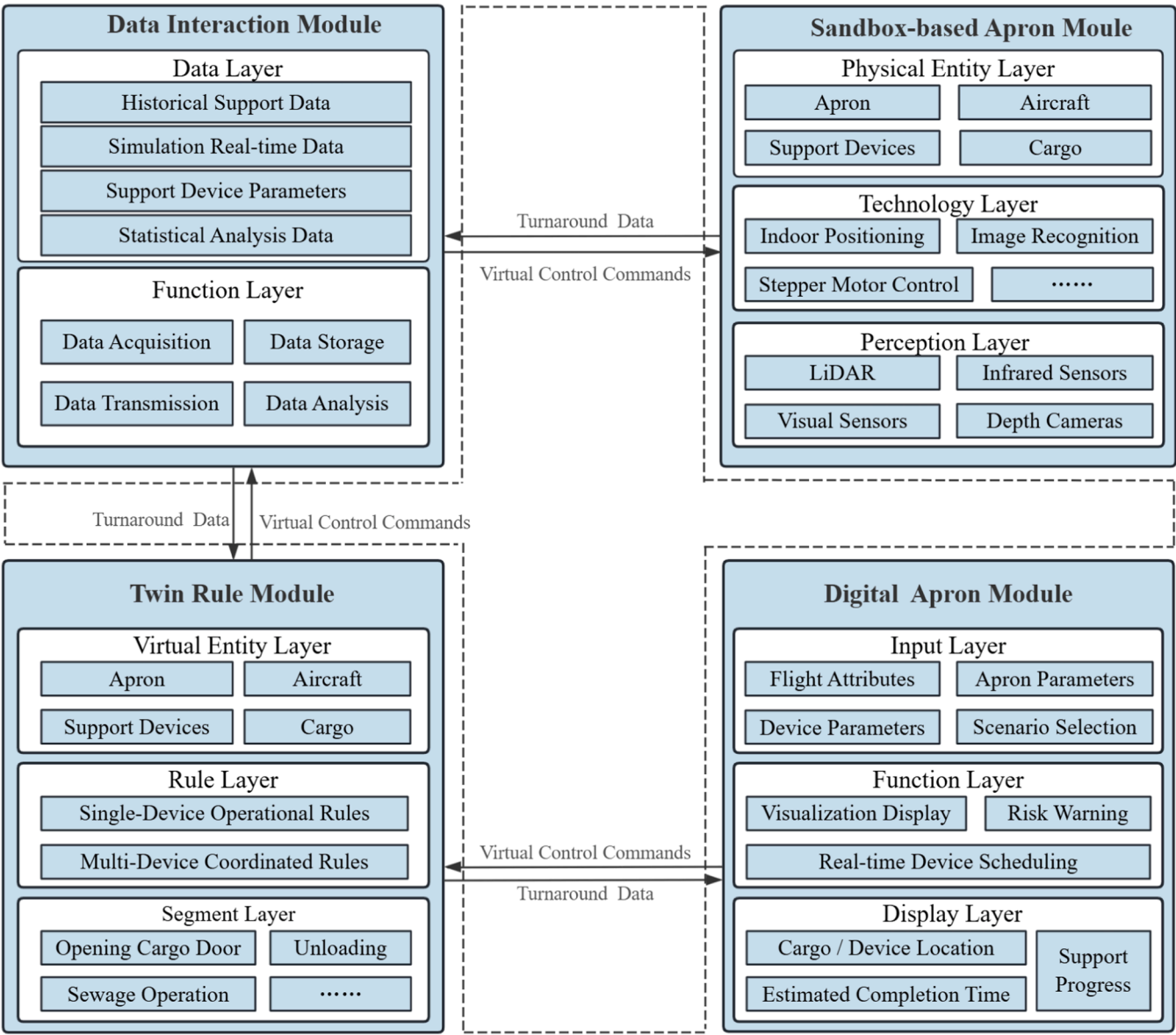


Fig. 1. Architecture of the simulation model.

3. Digital twin model for automated aircraft turnaround operation

Referring to the five-dimensional structure of the digital twin model proposed by Tao et al. [34], We conduct the simulation model of aircraft turnaround operation based on digital twin technology which requires the support of both computer hardware and software technologies. The specific parameters of the computer hardware are as follows: the simulation is supported by computer with independent graphics card 3060 (12GB/NVIDIA), 32 GB of memory, and 12th Gen Intel Core i7–12700F processor.

The architecture of the simulation model can be seen in Fig. 1. The simulation model consists of four modules: sandbox-based apron module, digital apron module, data interaction module and twin rule

module. Within the structure, the sandbox-based apron module is used to construct the scaled-down smart apron in the laboratory, which is composed of the physical layer, perception layer and technology layer. The digital apron module serves as the virtual representation of the sandbox-based apron, which consists of the input layer, function layer and display layer. The data interaction module links the sandbox-based apron module and the digital apron module, enabling bidirectional transmission of communication and control data. The twin rule module is responsible for setting up the aircraft turnaround operation rules for various smart devices, including operational rules for single device as well as the ones for multi-device coordination.

The reminder of this section will put more focus on the development of the sandbox-based apron and the twin rule module.

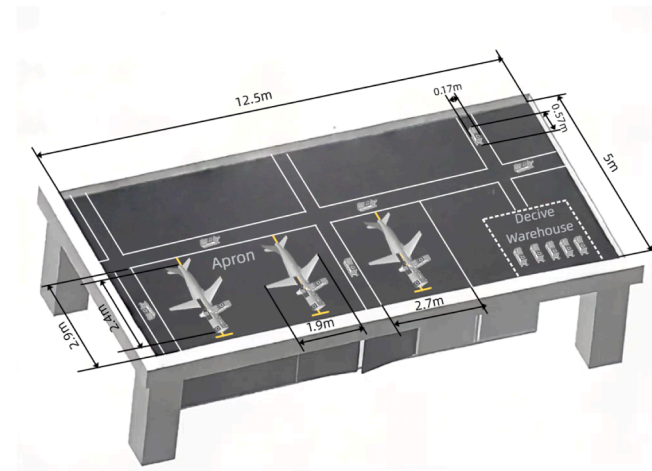


Fig. 2. The structural diagram of the sandbox.

Table 1
The motions and max-speed of devices in the sandbox-based apron (cm/s).

Device	Advancing	Reversing	Raising	Lowering
Aircraft	12	7.5	-	-
Cargo tractor	37	12	-	-
Conveyor belt vehicle	63	12	0.9	0.9
Lift Platform vehicle	63	12	0.9	0.9
Refueling vehicle	63	12	-	-
Sewage vehicle	63	12	-	-

3.1. Sandbox-based apron

In the sandbox, the aircraft is composed of the fuselage, wings, tail, landing gear, cargo hold doors, interior roller structure of the cargo hold. The smart devices consist of the automated and semi-automated

conveyor vehicles, lift platform vehicles, refueling vehicles, sewage vehicles, cargo tractors, etc. Cargos and cargo related devices involve trailers, containers, pallets, cargo nets, suitcases and bulk cargo.

As the smart apron has not appeared yet, our designed sandbox-based apron follows the features of the real apron in terms of layout and scale, but its size is scaled down proportionally to meet the spatial requirements of the laboratory. Meanwhile, the scaled-down smart devices in the sandbox are also designed by referring to the characteristics of the real devices currently used in reality, involving the shape and moving speed.

To be specific, the size of the sandbox and devices is determined by the real world and scaled down according to the proportional parameter 15:1. The apron sandbox measures 12.5 m of length, 5 m of width and 1 m of height; the apron measures 2.9 m of length and 2.7 m of width; the aircraft measures 2.4 m of length, 1.9 m of wingspan and 0.74 m of height; the lift platform vehicle measures 0.57 m of length, 0.17 m of width and 0.1 m of height...Then, the structural diagram of the sandbox is shown in Fig. 2.

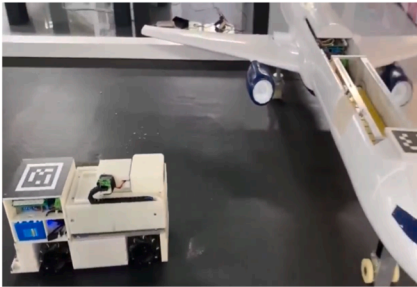
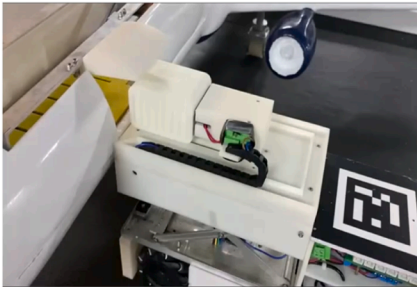
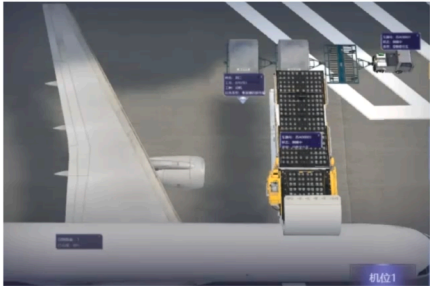
The design motions as well as speed of aircraft and the smart devices are shown in Table 1. From this table, we can see that the aircraft in the sandbox-based apron can conduct motions like opening and closing cargo door, positioning, and taxiing, etc. Similarly, the designed smart devices also are capable of achieving mechanical movements such as advancing, reversing, raising and lowering. It is worth nothing that the speed of different motion is determined by the electrically powered engine.

Considering space limitation, the lift platform vehicle is taken as an example to illustrate the sensor configurations on smart devices. The depth cameras and the LiDAR are used to collect vehicle location as well as its distance to the aircraft that to guarantee the precise location and obstacle avoidance. The encoder is installed on the wheel stepper motors to measure the vehicle speed and displacement. The temperature and pressure sensors are applied to monitor the state of hydraulic system, thus to prevent malfunctions caused by overheating or abnormal pressure. The micro camera can collect real-time visual data so as to assist path planning. The wireless communication part supports the data

Table 2
The operation workflow of the lift platform vehicle.

Step	Device			Hardware									Motion
	Lift platform vehicle	Aircraft	Cargo	PLC	Stepper motor	Wheel controller	Distance sensor	Speed sensor	Hydraulic system	Conveyor system	Pressure sensor	RFID tag	
1	√			√	√	√							Vehicle start-up Vehicle positioning Path planning and vehicle movement
2	√			√			√	√					
3	√			√	√	√	√	√					
4	√			√	√	√							Vehicle stop Loading cargo onto the platform vehicle Raise the platform and dock with the door
5	√		√	√						√	√	√	
6	√	√	√	√			√	√	√			√	
7	√	√	√	√			√	√		√	√	√	Loading the cargo into the cargo hold Lower the platform Vehicle start-up Vehicle positioning Path planning and Vehicle movement
8	√			√			√	√	√				
9	√			√	√	√							
10	√			√			√	√					Vehicle stop
11	√			√	√	√	√	√					
12	√			√	√	√							

Table 3
Selected operational images of the lift platform vehicle.

Segment	Sandbox apron	Digital apron
Docking with the aircraft		
Cargo transfer into the aircraft hold		
Cargo movement within the aircraft hold		

exchange between the physical lift platform vehicle and digital one, which facilitates the remote monitoring and dispatching.

The sandbox operates as a process in which the operational rules (see Section 3.2) can drive various smart devices to serve the flights automatically. In the simulation, when receiving the corresponding computer commands, the smart devices are driven by designed algorithms corresponding to the path planning, the stepper motor control and so on. Then they can collect and transmit the operational data through radar, infrared, visual sensors, etc. Meanwhile, the digital apron module conducts a virtual simulation based on the operational data.

Taking the lift platform vehicle as an example, we detail the specific workflow of the simulation in Table 2, which is added in the revised version. Owing to space limitation, Table 3 only presents selected operational images of the lift platform vehicle. The left side depicts the turnaround performance on the physical sandbox apron, while the right side shows the real-time operations in the digital apron.

3.2. Twin rule module

On the basis of the developed sandbox-based apron module, the digital apron module, and the data interaction module, the twin rule module primarily defines the operational rules of various smart devices. These operational rules drive the smart devices to serve the flights automatically. The operational rules consist of single-device operational rules and multi-device coordinated rules. The single-device operational

Table 4
The working segments of different devices.

Device	Segments
aircraft	park, cargo door open, cabin door open, cabin door close, cargo door close, release and pushback
cargo tractor	dock, unload, load
conveyor belt vehicle	dock, load, evacuate
lift platform vehicle	docking, baggage and cargo load, evacuate
refueling vehicle	dock, refuel, evacuate
sewage vehicle	dock, swage discharge, evacuate
aircraft tractor	dock, pushback, evacuate

rules are formulated based on the *Guidelines for the Configuration of Special Equipment for Civil Transport Airport Flight Support (Trial)* [3] and the *Flight Safety Operation Support Standards* [4], which define the detailed operational procedures of each device for different turnaround activities. The coordinated rules for multiple smart devices are mainly developed using the network planning technique to predict the enter and exit timing of different devices during the turnaround segment. To be specific, the segment refers to the steps evolved in the procedure of aircraft turnaround operation, such as opening the cargo door, loading/unloading and refueling, etc. Meanwhile, different from the segment construction for the turnaround operation with manual work,

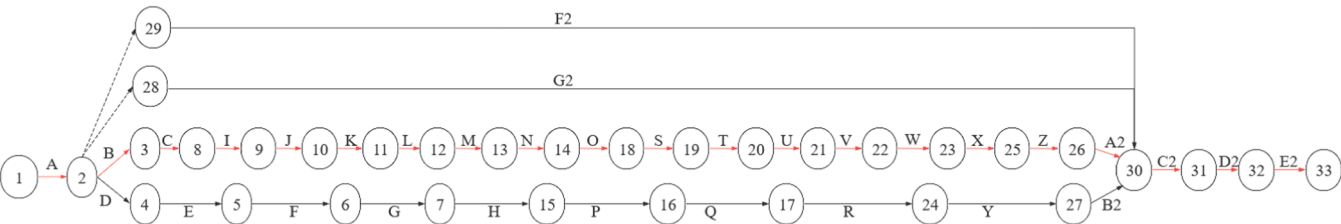


Fig. 3. Aircraft automated turnaround operation Network Diagram.

Table 5
The workflow for automated aircraft turnaround operation.

No.	Segment	Previous segment
A	Wheel chock and reflective cone placement	-
B	Opening the main cargo door	A
C	Docking the lift platform vehicle	B
D	Opening the lower cargo hold door	A
E	Docking the conveyor belt vehicle	D
F	Cargo tug a adjusts bulk cargo trailer position to dock with the conveyor belt vehicle	E
G	Unloading bulk cargo to the bulk cargo trailer	F
H	Departure of unmanned cargo tug a	G
I	Cargo tug b adjusts ULD (Unit Load Cargo) trailer position to dock with the lift platform vehicle	C
J	Unloading ULDs 1–3	I
K	Departure of unmanned cargo tug b	J
L	Cargo tug c adjusts ULD trailer position to dock with the lift platform vehicle	K
M	Unloading ULDs 4–6	L
N	Departure of unmanned cargo tug c	M
O	Unmanned cargo tugs e and f transport cargo to the stand-by position	N
P	Cargo tug d adjusts bulk cargo trailer position to dock with the conveyor belt vehicle	O
Q	Loading bulk: cargo into the lower cargo hold	P
R	Departure of unmanned cargo tug d	Q
S	Cargo tug e adjusts ULD trailer position to dock with the lift platform vehicle	O
T	Loading ULDs 1–3	S
U	Departure of unmanned cargo tug e	T
V	Cargo tug f adjusts ULD trailer position to dock with the lift platform vehicle	U
W	Loading ULDs 4–6	V
X	Departure of unmanned cargo tug f	W
Y	Withdrawal of the conveyor belt vehicle	R
Z	Withdrawal of the lift platform vehicle	X
A2	Closing the main cargo door	Z
B2	Closing the lower cargo hold door	Y
C2	Docking the towing tractor	A2, B2
D2	Removal of wheel chocks and reflective cones	C2
E2	Aircraft pushback	D2
F2	Sewage operations	-
G2	Refueling operations	-

the features of smart devices are particularly concerned in the segment construction of autonomous turnaround operation. For example, the segments conducted by the smart tug involve arriving at the waiting position, docking with the conveyor belt vehicle, loading cargo and evacuation.

3.2.1. Single-device operational rules

The single-device operational rules consist of the workflow rules and

movement regulations. The workflow covers the segments involving docking to the aircraft, performing the operation, and traveling back to the parking place. The working segments of different devices are shown in Table 4. In the table, the aircraft has the most complicated workflow involving six segments, but the cargo tractor has the simplest one, which is composed of two segments.

Different from the manual-device coordinated apron, the smart devices can serve the flights automatically in the sandbox-based apron under the digital control. For instance, the cargo tractors autonomously receive tasks from the cloud, follow the pre-planned route, transfer the cargo to the elevator platform vehicle and so on.

Besides the workflow, devices should also follow the movement regulation. They can be explained from the following perspectives: speed, distance, and angle. The travel speed must not exceed 0.055 m/s, and the docking speed is generally below 0.04 m/s. If device uses a telescopic platform to dock with the aircraft, the extension speed of the platform should not exceed 0.004 m/s. After the docking is completed, the distance between the vehicle and the aircraft cargo door is within the range of 0.2–0.4 centimeters for the lift platform vehicle and conveyor belt. The pitch angle of the conveyor belt vehicle is less than 24 degrees.

3.2.2. Multi-device coordinated rules

The fundamental issue of constructing the multi-device coordinated rules is to set the entering or exiting time of different smart devices for the aircraft turnaround operation, and further to plan the autonomous trajectories as well as movement details of each smart device. As our research focuses on the autonomous turnaround operation of a single flight and the surface area of sandbox is enough for the movements of smart devices, the planning of autonomous trajectory for smart devices is relatively easy. In the sense, the multi-device coordinated rules in the paper primarily refer to the specific timing of smart devices when entering or exiting the autonomous turnaround operation.

Such coordinated rules are established based on the network planning technique in Operations Research, the specific steps as follows: (1) depict the process of autonomous turnaround operation, as shown in Fig. 3; (2) collect the average duration spent by different smart devices in performing a certain task based on the digital twin technology, as shown in Table 7; (3) calculate the enter and exit time of smart devices using the network planning techniques, so as to minimize the duration of the aircraft turnaround operation, as presented in Eqs. (1)–(6). On the basis of above steps, the coordinated rules of smart devices are further established. Further details are as follows.

First, Table 5 gives out all the segment during the automated turnaround operation.

Then, the whole procedure can be depicted by the network plan shown in Fig. 3. In the figure, the circle represents the time point and the arrow represents the segment in Table 5. The start segment is wheel chock and reflective cone placement, the end segment is aircraft

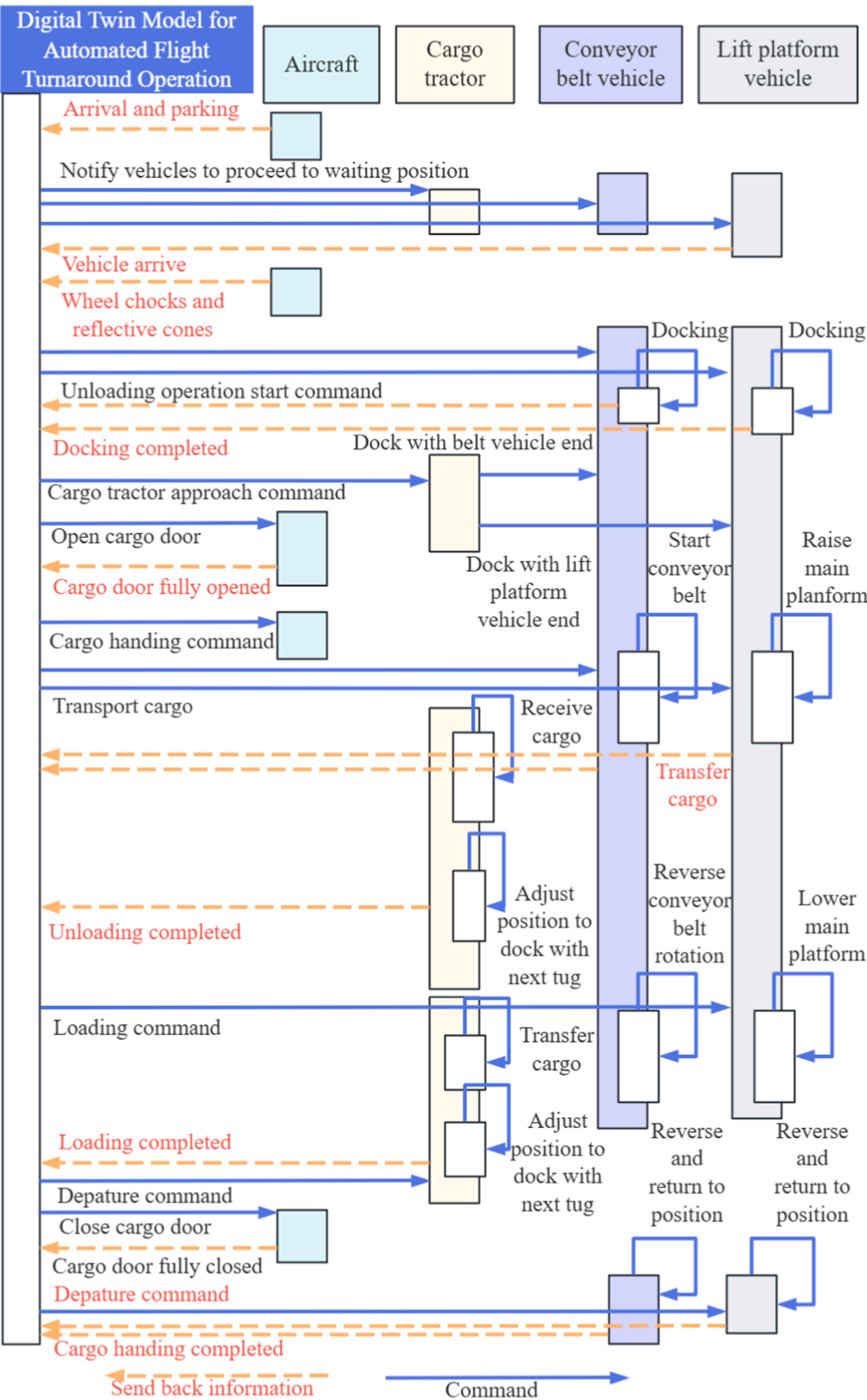


Fig. 4. Interaction between the digital apron module and the sandbox-based apron.

Table 6

Collected data about the apron and device in reality.

Device	Size			Operating parameters
	Length/ m	Width/ m	Height/ m	
Aprons	53.5	43	-	-
Aircraft	36.5	28.9	11.1	-
Cargo Tractor	3.15	1.44	1.45	Maximum towing weight: 38000 kg Maximum traveling speed: 29 km/h
Conveyor Belt Vehicle	8.09	2	1.64	Transfer speed: 0.2 m/s Maximum load ability on average: 135 kg/m
Lift Platform Vehicle	8.62	3.7	3.04	Maximum loading weight: 68,000 kg Maximum lifting height: 3.5 m
Refueling Vehicle	14.02	3.05	3.0	Volume: 37,800 L Flow rate: 38 L/s
Sewage Vehicle	6.55	1.9	2.15	Volume: 2000 L Flow rate: 2.5 L/s
Aircraft Tractor	5.08	2.2	1.88	Rated traction: 250,000 kg

Table 7

Operational time for the automated turnaround.

Segment	Time interval/s	Segment	Time interval/s
A	180	R	29
B	58	S	51.5
C	52.5	T	200
D	16	U	20
E	50	V	35
F	42	W	198.5
G	349.5	X	25
H	33.5	Y	50
I	25	Z	53.5
J	136	A2	41
K	26	B2	11
L	20.5	C2	180
M	120	D2	180
N	29	E2	180
O	45	F2	900
P	65.5	G2	720
Q	275.5		

pushback, and the order of other segments is set according to the designed workflow. It needs to be pointed out that the segments in the procedure are not simply in tandem connection but in a parallel manner, so the network planning technique is applied to determine the enter and exit times of each segment conducted by different smart devices.

The procedure of determining the start and finish times is as follows. First, we should calculate the total earliest start time and finish time of the automated aircraft turnaround operation, which is calculated using Eqs. (1) to (3). In the Equations, i, j, k denotes the time nodes, which belong to the set I . Second, the latest start times and finish times of each segment are calculated using Eqs. (4) to (6).

$$t_{ES}(1, j) = 0 \quad (1)$$

$$t_{ES}(i, j) = \max_k \{t_{ES}(k, i) + t_{ki}\} \quad (2)$$

$$t_{EF}(i, j) = t_{ES}(i, j) + t_{ij} \quad (3)$$

$$t_{LF}(i, n) = t_{EF}(i, n) \quad (4)$$

$$t_{LF}(i, j) = \min_k \{t_{LF}(j, k) - t_{jk}\} \quad (5)$$

$$t_{LS}(i, j) = t_{LF}(i, j) - t_{ij} \quad (6)$$

Table 8

Operational time for the manual-device turnaround.

Segment	Time interval/ s
A3	Wheel chock and reflective cone placement
B3	Cargo door opening
C3	Loading /unloading
D3	Sewage operation
E3	Refueling
F3	Cargo door closing
G3	Tug connection
H3	Wheel chock and reflective cone removal
I3	Aircraft pushback

Then we should find out the key path on which the earliest and latest start times of every segment are equal to each other, and the enter time of the smart device for the segments on this key path should be strictly determined as t_{ES} in Eq. (2). Besides, the enter time of the smart device for the segments which are not on the key path should be set as a random value within the range $[t_{LS} - t_{ES}]$. With the determined start and finish times of each segment, we can construct the coordinated rules for every smart device in the smart apron.

Third, with the designed rules for single-device and multi-device coordination, this study has implemented the interaction between the digital apron module and the sandbox-based apron by applying multi-device collaboration rules, as shown in Fig. 4. In the process, the digital apron module sends commands such as entering, docking, and withdrawing to the corresponding scaled-down devices according to the requirements of the operation process. After receiving the commands, the devices perform the equivalent operations and return the completion signals to the digital apron module, which records the duration of the segments.

4. Case study

In this section, we use the data from one specific airport in the southeast part of China to validate the digital-twin model in Section 3, especially for constructing the sandbox-based apron.

4.1. Data collection

The collected data are composed of two parts. The first part involves the data about the size, power and speed related to the apron, aircraft and devices actually used in reality, which are depicted in Table 6. In the table, the scale of the apron covers 2300 square meters, and the operating parameters of each device in the manual-device coordinated way are also collected according to the statistical data provided by the airport.

Table 7 depicts the predicted duration for each segment with smart devices. These predicted values represent the average of 20 simulation results based on the digital twin model constructed in our research. Additionally, these flights primarily transport unit load cargo and loose cargo, with unit load cargo loaded into the main hold and loose cargo placed in the belly hold.

The procedure of the turnaround operation with manual work is relatively easier, while the autonomous turnaround operation with smart devices needs to be driven by control rules related to all movements of smart devices. Hence, the volume of segments in Table 7 is more than that of Table 8.

4.2. Prediction of the efficiency of aircraft automated turnaround operation

With the constructed digital-twin model and the collected data, the start and finish times of every segment are set according to the calculated network plan shown in Fig. 5. In the figure, the earliest start time of the first segment is set to be 0, and the earliest finish-time of the whole

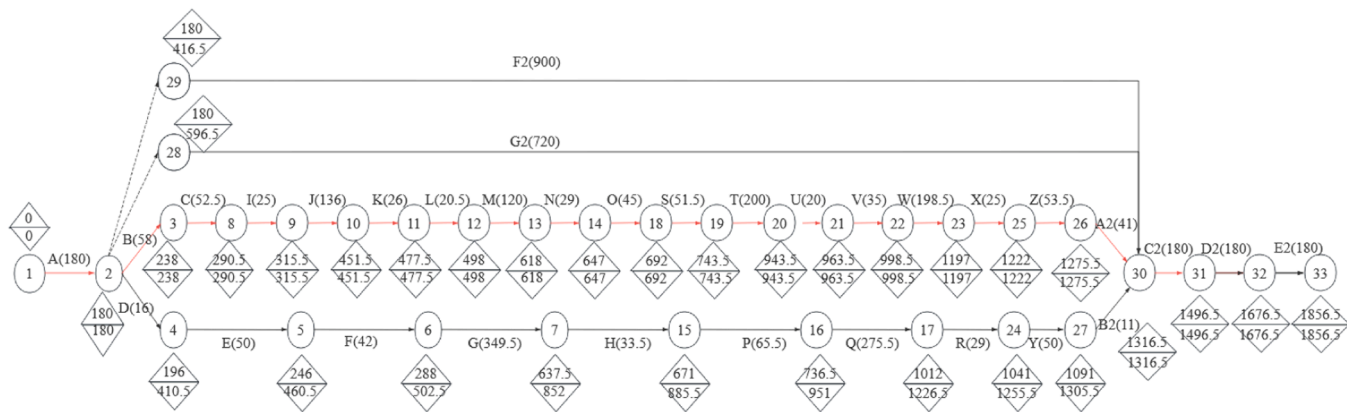


Fig. 5. Aircraft automated turnaround operation AOE network.

Table 9
The start and finish time of each segment.

Segment	Start time/s	Finish time/s	Segment	Start time/s	Finish time/s
A	0	180	R	1012–1226.5	1041–1255.5
B	180	238	S	692	743.5
C	238	290.5	T	743.5	943.5
D	180–394.5	196–410.5	U	943.5	963.5
E	196–410.5	246–460.5	V	963.5	998.5
F	246–460.5	288–502.5	W	998.5	1197
G	288–502.5	637.5–852	X	1197	1222
H	637.5–852	671–885.5	Y	1041–1255.5	1091–1305.5
I	290.5	315.5	Z	1222	1275.5
J	315.5	451.5	A2	1275.5	1316.5
K	451.5	477.5	B2	1091–1305.5	1316.5
L	477.5	498	C2	1316.5	1496.5
M	498	618	D2	1496.5	1676.5
N	618	647	E2	1676.5	1856.5
O	647	692	F2	180–416.5	1316.5
P	671–885.5	736.5–951	G2	180–596.5	1316.5
Q	736.5–951	1012–1226.5			

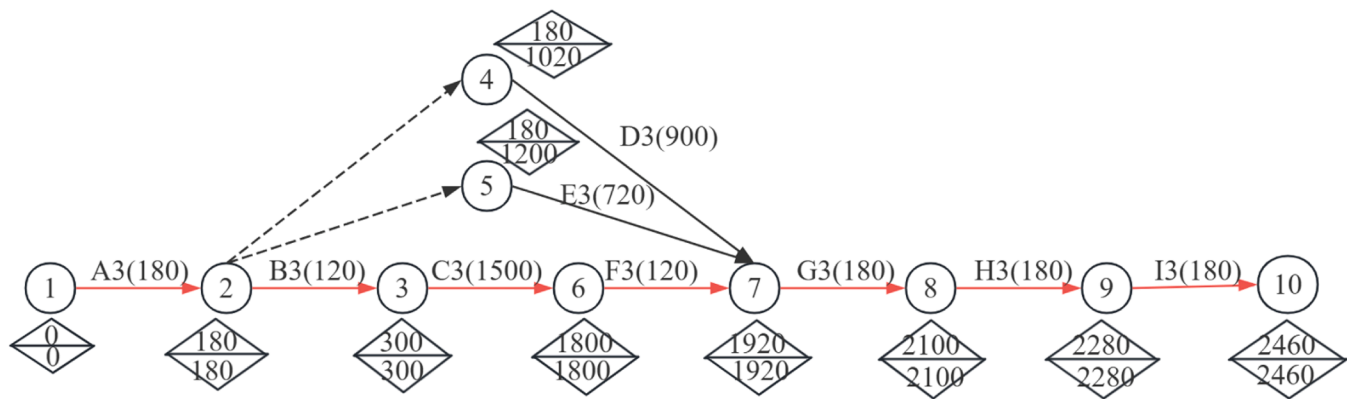


Fig. 6. The aircraft manual-device coordinated turnaround operation AOE network.

automated aircraft turnaround operation is 1856.5 s. Besides, the key path of the procedure is $A \rightarrow B \rightarrow C \rightarrow I \rightarrow J \rightarrow K \rightarrow L \rightarrow M \rightarrow N \rightarrow O \rightarrow S \rightarrow T \rightarrow U \rightarrow V \rightarrow W \rightarrow X \rightarrow Z \rightarrow A2 \rightarrow C2 \rightarrow D2 \rightarrow E2$, and the start and finish times of each segment are set in Table 9. The final duration of the automated aircraft turnaround operation is predicted to be 1856.5 s.

Next, in order to compare the efficiency of the manual-device coordinated operation and the automated one, the statistical operational duration in the actual apron is calculated to be 2460 s. The relevant planning network is shown in Fig. 6. The key path is $A3 \rightarrow B3 \rightarrow C3 \rightarrow F3 \rightarrow G3 \rightarrow H3 \rightarrow I3$.

The comparison indicates that the application of automation technology significantly improves the efficiency of cargo aircraft turnaround operation, reducing total duration by 24.53 %. By comparing and analyzing the duration of aircraft turnaround operation supported by manual work as well as smart devices (see Figs. 5 and 6), we summarize the advantages of the autonomous turnaround operation as follows:

- (1) Reduction in the duration of single segment. Specifically, for critical segments which significantly influence the overall turnaround efficiency (e.g., cargo loading/unloading, opening/closing the cargo doors), the duration served by smart devices is significantly shorter than that of manual operation.
- (2) Refinement of turnaround procedure and reduction in the connection duration between consecutive segments. The segments involved in the autonomous turnaround operation are refined according to the control rules of smart devices (see Table 5). As a result, two consecutive segments can be conducted by different smart devices without delay, thus to effectively reduce the coordination time between segments. In the contrast, the procedure of the turnaround operation with manual work may not be constructed in detail (Table 8), the performance of specific segment is highly depended on workers' proficiency, resulting in less efficiency.

In the sense, the reduced duration of the automated turnaround operation is not attributed to the omission of specific segments, but rather to the refinement of segment construction as well as the systematic implementation of device operation rules.

5. Conclusion

In the context of continuously increasing flight numbers and growing shortage of labor force, this study aims to explore the impact of smart devices on enhancing the efficiency of aircraft turnaround operation. In this sense, we develop the digital twin model integrating the aircraft automated turnaround operation and conduct a case study. Due to the lack of an automated apron in reality, we develop a scaled-down sandbox-based apron, which is equipped with scaled-down aircraft and smart devices. By setting single-device operational rules and multi-device coordinated rules, the sandbox devices can be driven to simulate actual aircraft turnaround processes and predict the operational efficiency. In the model construction, the network planning technique is developed to analyze the key paths and the total time duration of automated and manual-device operation respectively. The results indicate that the duration of automated operation can be reduced by

24.53 % compared to the manual-device coordinated operation.

Meanwhile, our proposed model not only validates the significant role of smart devices in improving the efficiency of aircraft turnaround operations, but also provides a theoretical practical reference for the application of digital twin technology in the field of airport management, especially for risk early warning and real-time scheduling. However, due to the strict safety regulations, the smart devices in our research can only be applied in a laboratory environment and not in reality. Considering the discrepancies between sandbox data and real operational data, our predicted efficiency of automated turnaround operation may not be that precise, but the results can be used as a reference for airport managers.

Future work will focus on enhancing the practicality and flexibility of our model in real-world environments, with the objective of reducing the influence and constraints imposed by human factors on operational efficiency. That includes intensifying the investigation into the interplay between personnel and machinery to refine collaborative processes, and bolstering the mechanisms for real-time monitoring and operational adjustments to guarantee punctuality and precision.

CRediT authorship contribution statement

Jiaxin Ji: Writing – original draft, Writing – review & editing, Methodology. **Xuheng Lu:** Conceptualization, Validation, Writing – original draft, Writing – review & editing. **Tao Pang:** Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Jing Lu:** Funding acquisition, Methodology, Project administration, Resources, Software, Supervision, Validation, Writing – original draft, Writing – review & editing. **Changmin Jiang:** Methodology, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Changmin Jiang reports financial support was provided by National Natural Science Foundation of China. Jing Lu reports financial support was provided by National Natural Science Foundation of China. Changmin Jiang reports financial support was provided by the Research Center for Low-Altitude Economy (RCLAE), the Hong Kong Polytechnic University. Jing Lu reports financial support was provided by 2024 National Foreign Expert Program of China. Given his role as editor, Changmin Jiang had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table 10

Result of twenty simulation experiments (Time interval / s)

No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A	179.0	180.5	181.0	179.5	178.5	180.0	180.5	179.0	181.5	180.0	178.5	180.0	179.5	180.5	181.0	179.0	180.0	180.5	179.5	180.0
B	57.5	58.0	57.0	58.5	58.0	57.5	57.0	58.0	59.0	58.5	57.5	58.0	57.5	57.0	58.5	58.0	58.0	57.5	58.5	58.0
C	52.0	52.5	53.0	52.0	52.5	53.5	51.5	52.5	52.0	52.5	53.0	52.5	52.0	52.5	53.0	51.5	52.5	52.0	52.5	53.5
D	15.5	16.0	16.0	15.5	16.0	16.5	16.0	15.5	16.0	16.5	16.0	15.5	16.0	16.0	15.5	16.5	16.0	16.0	15.5	16.0
E	50.0	49.5	50.5	49.0	50.0	50.5	49.5	51.0	50.0	49.5	50.0	50.0	49.5	50.5	50.0	49.0	50.5	50.0	49.5	50.5
F	41.0	42.5	42.0	41.5	42.0	41.0	42.5	42.0	43.0	41.5	42.0	42.5	41.0	42.0	42.5	41.5	42.0	41.5	43.0	42.0
G	348.5	350.0	349.0	350.5	349.5	348.0	349.5	350.0	348.5	350.5	349.0	350.0	348.5	349.5	350.0	348.5	349.0	350.5	349.5	349.0
H	33.5	34.0	33.0	33.5	34.0	33.0	33.5	34.0	33.5	33.0	34.0	33.5	33.0	33.5	34.0	33.5	33.0	34.0	33.5	33.5
I	25.0	24.5	25.0	25.5	24.5	25.0	25.5	25.0	24.5	25.0	25.5	25.0	24.5	25.0	25.5	25.0	24.5	25.0	25.5	25.0
J	136.0	135.5	136.5	136.0	136.5	135.0	136.0	137.0	135.5	136.0	136.5	135.0	136.5	136.0	135.5	136.0	137.0	135.5	136.0	136.5
K	26.0	25.5	26.5	26.0	25.5	26.5	26.0	26.0	25.5	26.5	26.0	25.5	26.5	26.0	26.0	25.5	26.5	26.0	26.0	25.5
L	20.0	20.5	21.0	20.5	20.0	21.0	20.5	20.0	21.0	20.5	20.0	21.0	20.5	20.0	20.5	21.0	20.5	20.0	20.5	21.0
M	119.5	120.5	121.0	119.0	120.0	120.5	119.5	120.0	121.5	119.0	120.5	120.0	119.5	120.0	120.5	119.0	121.0	120.0	119.5	120.0
N	28.5	29.0	29.5	29.0	28.5	29.0	29.5	29.0	28.5	29.0	29.5	29.0	28.5	29.0	29.5	29.0	28.5	29.0	29.5	29.0
O	45.0	44.5	45.5	45.0	44.5	45.0	45.5	45.0	44.5	45.0	45.5	45.0	44.5	45.0	45.5	45.0	44.5	45.0	45.5	45.0
P	65.0	65.5	66.0	65.0	66.0	65.5	65.0	66.5	65.0	65.5	66.0	65.0	65.5	66.0	65.0	65.5	66.0	65.0	65.5	66.0
Q	275.0	276.0	275.5	274.5	276.0	275.5	275.0	276.5	275.0	275.5	276.0	275.0	275.5	274.5	276.0	275.5	276.0	275.0	275.5	276.0
R	29.0	28.5	29.0	29.5	29.0	28.5	29.0	29.5	29.0	28.5	29.0	29.5	29.0	28.5	29.0	29.5	29.0	28.5	29.0	29.5
S	51.0	52.0	51.5	52.0	51.0	52.5	51.5	51.0	52.0	51.5	52.0	51.0	52.5	51.5	51.0	52.0	51.5	52.0	51.0	52.5
T	200.0	199.5	200.5	201.0	199.0	200.5	200.0	199.5	201.0	199.5	200.0	200.5	199.0	200.5	200.0	199.5	200.5	199.0	201.0	200.0
U	20.0	19.5	20.5	20.0	19.5	20.5	20.0	20.5	19.5	20.0	20.5	19.5	20.0	20.5	19.5	20.0	20.5	19.5	20.0	20.5
V	35.0	34.5	35.5	35.0	35.0	34.5	35.5	35.0	34.5	35.0	35.5	35.0	34.5	35.0	35.5	35.0	34.5	35.0	35.5	35.0
W	198.0	198.5	199.0	198.0	199.5	198.5	198.0	199.0	198.5	198.0	199.5	198.5	198.0	199.0	198.5	198.0	199.0	198.5	199.5	198.5
X	25.0	25.0	24.5	25.5	25.0	24.5	25.5	25.0	25.0	24.5	25.5	25.0	25.0	24.5	25.5	25.0	25.0	24.5	25.5	25.0
Y	50.0	49.5	50.5	50.0	50.0	49.5	50.5	50.0	50.0	49.5	50.5	50.0	50.0	49.5	50.5	50.0	50.0	49.5	50.5	50.0
Z	53.5	53.0	54.0	53.5	53.0	54.0	53.5	53.0	54.0	53.5	53.0	54.0	53.5	53.0	54.0	53.5	53.0	54.0	53.5	53.5
A2	41.0	40.5	41.5	41.0	40.5	41.5	41.0	41.0	40.5	41.5	41.0	41.0	40.5	41.5	41.0	41.0	40.5	41.5	41.0	41.0
B2	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0	11.0
C2	180.0	179.5	180.5	180.0	180.5	179.0	180.0	181.0	179.5	180.0	180.5	179.5	180.0	180.5	179.0	181.0	180.0	179.5	180.5	180.0
D2	180.0	179.5	180.5	181.0	179.0	180.5	180.0	179.5	181.0	180.0	179.0	181.5	180.0	179.5	180.5	180.0	180.0	179.5	180.5	180.0
E2	180.0	180.5	181.0	179.5	180.0	179.0	180.5	180.0	179.5	180.5	180.0	181.0	179.0	180.0	180.5	179.5	180.0	180.5	180.0	179.5
F2	900.0	902.5	899.0	901.0	903.0	898.5	900.5	899.5	901.5	902.0	897.5	900.0	903.0	898.0	902.0	899.5	901.5	900.0	899.0	903.0
G2	720.0	721.0	719.5	720.5	718.5	721.0	720.0	719.0	722.0	720.5	719.0	721.5	720.0	719.5	720.5	718.5	721.0	720.0	719.5	720.5

Data availability

The authors do not have permission to share data.

References

- [1] Z. Aydın, O. Yardımcı, Regulatory sandboxes and pilot projects: trials, regulations, and insights in energy transition, *Eng. Sci. Technol.* 56 (2024) 101792, <https://doi.org/10.1016/j.jestch.2024.101792>.
- [2] R. Basu, R. Ponce-Lopez, J. Ferreira, Framework to generate virtual cities as sandboxes for land use-transport interaction models, *J. Transp. Land Use* 14 (1) (2021) 303–323. (<https://www.jstor.org/stable/48646187>).
- [3] Civil Aviation Administration of China Guidelines for the configuration of special equipment for civil transport airport flight support (trial). <http://www.caac.gov.cn/big5/www.caac.gov.cn/PHONE/XXGK/17/XXGK/GFXWJ/201606/P020160601354094286348.pdf>2015.
- [4] Civil Aviation Administration of China The flight safety operation support standards. <http://www.caac.gov.cn/XXGK/XXGK/ZFGW/202003/P020200319625809747758.pdf>2020.
- [5] Y. Cui, L. Ma, Q. Ding, et al., Aircraft turnaround time dynamic prediction based on time transition petri net, *PLOS One* 1;2006 (2024), <https://doi.org/10.1371/journal.pone.0305237>.
- [6] Y. Di, 2019, Behind digital intelligence and user-friendly services, how are smart airports affecting people's travelling. *Interface News*. <https://www.jiemian.com/article/3763783.html>.
- [7] J. Ding, Y. Wu, A dynamic prediction model of flight chain delay based on digital twin, *J. Nanjing Univ. Aeronaut. Astronaut.* 55 (05) (2023) 859–867, <https://doi.org/10.16356/j.1005-2615.2023.05.012>.
- [8] J. Fan, X. Zhu, W. Gao, et al., Carrier aircraft turnaround support operation scheduling based on parallel genetic algorithm, *J. Ordnance Equip. Eng.* (2019). (https://en.cnki.com.cn/Article_en/CJFDTotol-CUXI201911028.htm).
- [9] Guo, Y. 2024. "Flying" out of high-quality development "acceleration" Haikou Meilan International Airport to build domestic civil aviation benchmarking services and world-class service brands. Civil Aviation Administration of China Website. (https://mp.weixin.qq.com/s?_biz=MjM5ODk5MDg2Mw==&mid=2654240044&idx=1&sn=9350be5ca5664b259a4a2f939d40704f&chksm=bc0dc0089f985c61340b0d005049a5576672419d5c78b6e286693b3ac9b7d96e35956d88f1bc&scene=27).
- [10] L. Hu, Z. Liu, J. Tan, A VR simulation framework integrated with multisource CAE analysis data for mechanical equipment working process, *Comput. Ind.* 97 (2018) 85–96, <https://doi.org/10.1016/j.compind.2018.01.009>.
- [11] T. Huang, Trajectory planning for unmanned vehicles on airport apron under aircraft–vehicle–airfield collaboration, *Sensors* 25 (2024), <https://doi.org/10.3390/s25010071>.
- [12] H. Jiang, S. Qin, J. Fu, et al., How to model and implement connections between physical and virtual models for digital twin application, *J. Manuf. Syst.* 58 (2021) 36–51, <https://doi.org/10.1016/j.jmsy.2020.05.012>.
- [13] R. KanableSmart solution manages aircraft turnaround activities from landing to takeoff. *Ground Support Worldwide*, 31(8), pp. 32–33. <https://blog.adbsafegate.com/smart-solution-manages-aircraft-turnaround-activities-from-landing-to-takeoff/2023>.
- [14] T. Kong, T. Hu, T. Zhou, et al., Data construction method for the applications of workshop digital twin system, *J. Manuf. Syst.* 58 (2021) 323–328, <https://doi.org/10.1016/j.jmsy.2020.02.003>.
- [15] S. Latif, J. Ferzund, Smart airport apron management system formal modeling using vdm-sl. *Proceedings of the 2019 13th International Conference on Mathematics, Actuarial Science, Computer Science and Statistics*, 2019.
- [16] E. Leckenby, D. Dawoud, J. Bouvy, et al., The sandbox approach and its potential for use in health technology assessment: a literature review, *Appl. Health Econ. Health Policy* 19 (2021) 857–869, <https://doi.org/10.1007/s40258-021-00665-1>.
- [17] Y. LiuAsia's first fully automated boarding bridge put into operation at Tianfu Airport. The People's Government of Sichuan Province. <https://www.sc.gov.cn/10462/10464/11716/11718/2022/8/6/ba87d1ec65ce4b37b9e61e64b69be1c2.shtml>2022.
- [18] X. Liu, D. Jiang, B. Tao, et al., A systematic review of digital twin about physical entities, virtual models, twin data, and applications, *Adv. Eng. Inform.* 55 (2023) 101876, <https://doi.org/10.1016/j.aei.2023.101876>.
- [19] R. Liu, Z. Wang, F. Liou, Multifeature-fitting and shape-adaption algorithm for component repair, *J. Manuf. Sci. Eng.* 140 (2) (2018) 021003, <https://doi.org/10.1115/1.4037107>.
- [20] W. Mao, 2024, Wulumuqi International Airport released 96.21 per cent of normal flights during the National Day period. China Daily Website. <http://ex.chinadaily.com.cn/exchange/partners/82/rss/channel/cn/columns/j3u3t6/stories/WS67053976a310b59111d9cf58.html>.
- [21] Mass Digital HealthThe digital health sandbox program. <https://www.massdigitalhealth.org/mass-digital-health-programs/digital-health-sandbox-program2020>.
- [22] D. Miltiadou, S. Pitsios, D. Spyropoulos, A secure experimentation sandbox for the design and execution of trusted and secure analytics in the aviation domain. In: *Proceedings of International Conference on Security and Privacy in New Computing Environments*2020.
- [23] V. Natarajan, R. Kumari, M. PriyaSystem and method for enhancing airlines turnaround operations. In: *Proceedings of the International Conference on Industry 4.0 and Advanced Manufacturing*. Doi: 10.1007/978-981-97-7150-9_152025.
- [24] National Bureau of StatisticsVice director of the national bureau of statistics answers questions on national economic performance for the first three quarters of 2024. Chinese Government Website. https://www.gov.cn/zhengce/202410/content_6981390.htm2024.
- [25] Y. Peng, 2020, China release: in 2019, china's civil aviation industry ranked second globally, achieving 80.68 million hours of safe flight in transport aviation. *NetEase*. <https://www.163.com/dy/article/F27NBJTU05346961.html>.
- [26] R. Phanden, P. Sharma, A. Dubey, A review on simulation in digital twin for aerospace, manufacturing and robotics, *Mater. Today. Proc.* (2021), <https://doi.org/10.1016/j.matpr.2020.06.446>.
- [27] M. Qi, X. Zhu, D. Yuan, et al., Modeling of the apron traffic system under the scenario of mixed operation of manned and unmanned vehicles, *Nav. Ship Electron. Eng.* (2025), <https://doi.org/10.3969/j.issn.1672-9730.2025.01.018>.
- [28] S. Saggarr, M. Tomasella, G. Cattaneo, Enhanced operational management of airport ground support equipment for better aircraft turnaround performance. In: *Proceedings of the 2021 Winter Simulation Conference (WSC)*. 2021.
- [29] F. Skultéty, N. Stalmáková, Pre-flight inspections of aircraft emergency equipment via RFID technology, *Transp. Res. Procedia* (2018), <https://doi.org/10.1016/j.trpro.2018.12.010>.
- [30] Z. Sng, R. Hansman, A petri net framework for the representation and analysis of aircraft turnaround operations, *ICAT Rep. Pap.* (2019). (<https://hdl.handle.net/1721.1/121164>).
- [31] R. Söderberg, K. Wärmeffjord, J. Carlson, et al., Toward a digital twin for real-time geometry assurance in individualized production, *CIRP Ann.* 66 (1) (2017) 137–140, <https://doi.org/10.1016/j.cirp.2017.04.038>.
- [32] D. Steininger, A. Kriegler, W. Pointner, Towards scene understanding for autonomous operations on airport aprons. In: *Proceedings of the Asian Conference on Computer Vision*2022.
- [33] D. Tabares, Correction: aircraft ground operations automation. The turnaround 2.0, *AIAA Aviat. Forum* (2023), <https://doi.org/10.2514/6.2023-3405.c1>.
- [34] F. Tao, W. Liu, M. Zhang, et al., Five-dimension digital twin model and its ten applications, *Comput. Integr. Manuf. Syst.* 25 (01) (2019) 1–18, <https://doi.org/10.13196/j.cims.2019.01.001>.
- [35] K. Wang, Y. Lee, S. Angelica, Digital twin design for real-time monitoring – a case study of die cutting machine, *Int. J. Prod. Res.* 59 (21) (2020) 6471–6485, <https://doi.org/10.1080/00207543.2020.1817999>.
- [36] G. Wu, Over 80% of the flights at hohhot airport are supported by the boarding bridge, *Civ. Aviat. Adm. China Newsp.* (2023). (https://m.thepaper.cn/baijiahao_24406427).
- [37] Yang, Y. (2022). Sanya Airport takes multiple measures to do a good job of supporting the Spring Festival. Civil Aviation Administration of China Website. (<https://www.caonline.cn/jichang/jchot/926619.html>).
- [38] G. Yoon, J. Chow, S. Rath, A simulation sandbox to compare fixed-route, flexiblerroute transit, and on-demand microtransit system designs, *Arxiv Prepr. Arxiv* (2021) 2109, <https://doi.org/10.1007/s12205-022-0995-3>.
- [39] W. ZhangBeijing Daxing International Airport begins supplying aviation fuel. Civil Aviation Administration of China Website. http://www.caacnews.com.cn/1/7/201905/t20190505_1272835_wap.html2019.
- [40] S. Zhang, X. Li, X. Yuan, et al., Optimizing the aircraft turnaround schedules: an improved sliding time Windows approach based on MILP and CP models, *Comput. Oper. Res.* (2024), <https://doi.org/10.1016/j.cor.2023.106433>.
- [41] X. Zhang, Y. Qiao, Y. Yang, et al., Smod: Scene-specific-prior-based moving object detection for airport apron surveillance systems, *IEEE Intell. Transp. Syst. Mag.* 15 (1) (2021) 58–69, <https://doi.org/10.1109/ITS.2021.3122926>.
- [42] W. Zhang, H. Zhu, Y. Sun, Status and development recommendations of airport aprons automated devices technology. In: *Proceedings of the 19th Annual China Intelligent Transportation Conference (ITSAC 2024)* 2024.
- [43] H. Zhao, J. Liu, H. Xiong, et al., 3D visualization real-time monitoring method for digital twin workshop, *Comput. Integr. Manuf. Syst.* 25 (06) (2019) 1432–1443, <https://doi.org/10.13196/j.cims.2019.06.011>.
- [44] J. Zhu, Air Transport Organization, Science Press, 2018. (<http://find.nuaa.edu.cn/#/searchList/bookDetails/255788>).